



Muscle tissue and vessels

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MUSCLE TISSUE

A unique feature of muscle cells (or fibers) with high content of proteins called *myofibrils*, which contract after stimulation of the muscle fibers. The myofibrils are composed of short protein threads, the *myofilaments*. These can be divided into thin *actin filaments* and thick *myosin filaments*.

Muscle tissue is divided into *smooth* and *striated muscle tissue* according to structure and function.

General features

- Less extracellular matrix
- High energy demand –*lot of mitochondria*
- Myofilaments
- High calcium demand – *sarcoplasmic reticulum, Ca^{2+} -pumps and channels*

Filaments

Thin filaments:

diameter: approx. 6-8 nm , length: approx. 1 μm

ACTIN PROTEINS: F, G

Thick filaments:

diameter: approx. 15 nm, length: kb. 1,5 μm

MYOSIN II. PROTEIN

all thick filaments are built up by 200-300 myosin II. molecules

The cytoplasm of muscle cells (=sarcoplasm) represents mainly actin and myosin II. proteins.

Actin and myosin proteins are also presented in the most cell types (e.g. migrating cells, phagocytic cells and glandular cells).

CLASSIFICATION OF MUSCLE TISSUES

STRIATED MUSCLE:

- SKELETAL MUSCLE
- VISCERAL STRIATED MUSCLE

(tongue, pharynx, upper part of the esophagus, lumbar part of the diaphragm) – speaking, swallowing and respiration

- CARDIAC MUSCLE

SMOOTH MUSCLE:

(wall of vessels and organs, internal eye muscles, arrector pili muscle)

SKELETAL MUSCLE

- Multi-nucleated cells
- The nuclei are located right under the cell membrane (*sarcolemma*)
- Length: from a few millimetres to one meter

SKELETAL MUSCLE

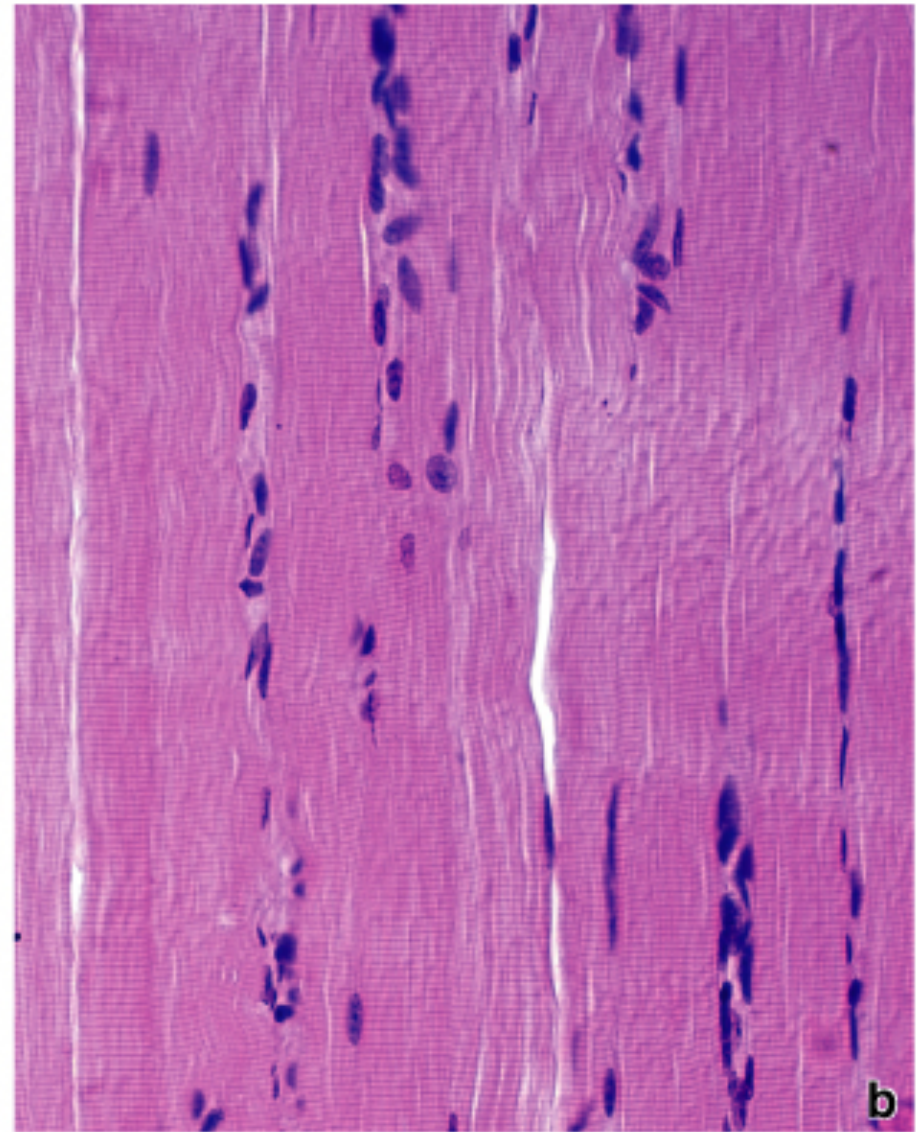
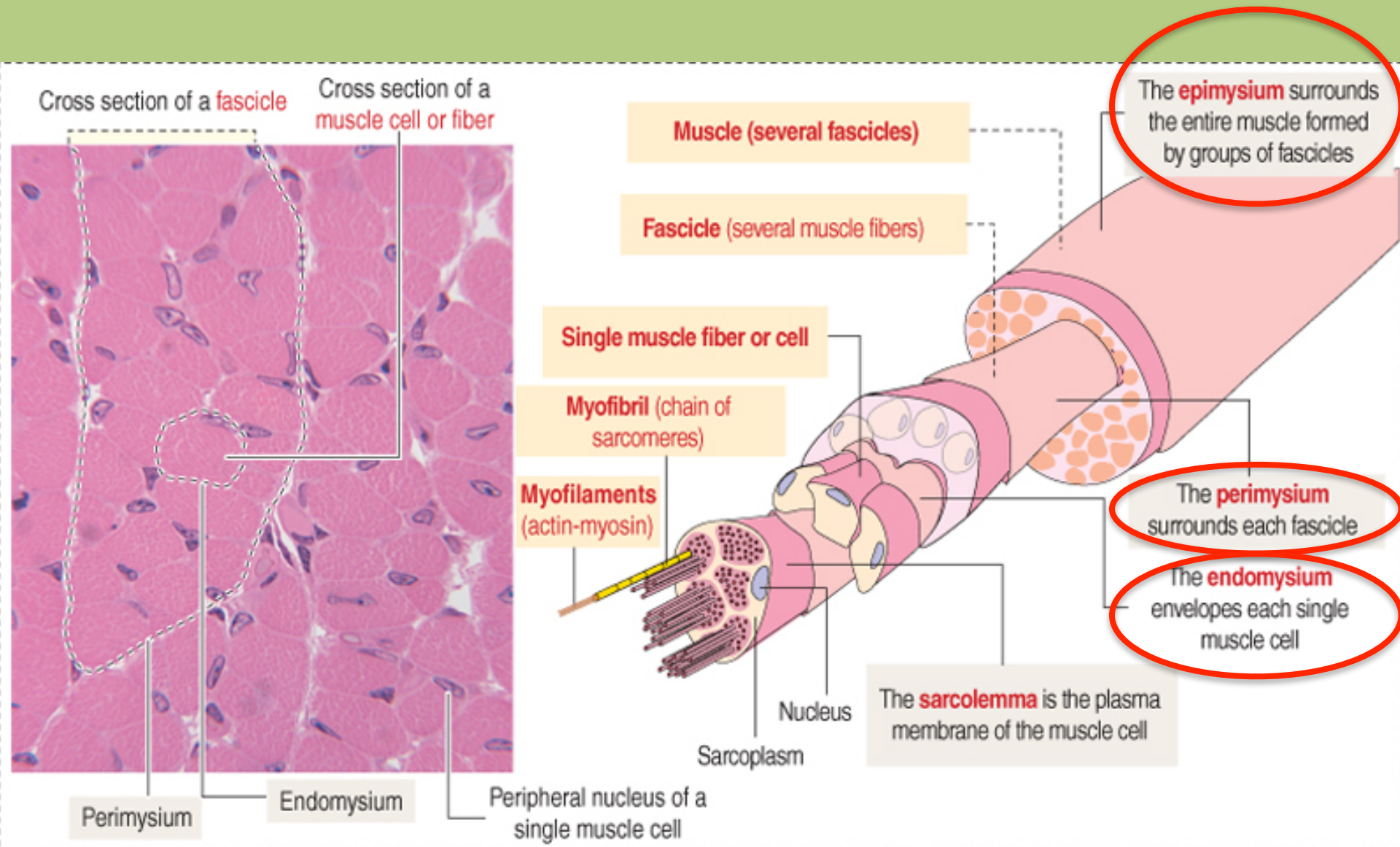


Figure 10.1b. Higher magnification photomicrograph of a skeletal muscle. X360.

Connective tissue coverings



STRIATION

A band: myosin

I band: actin

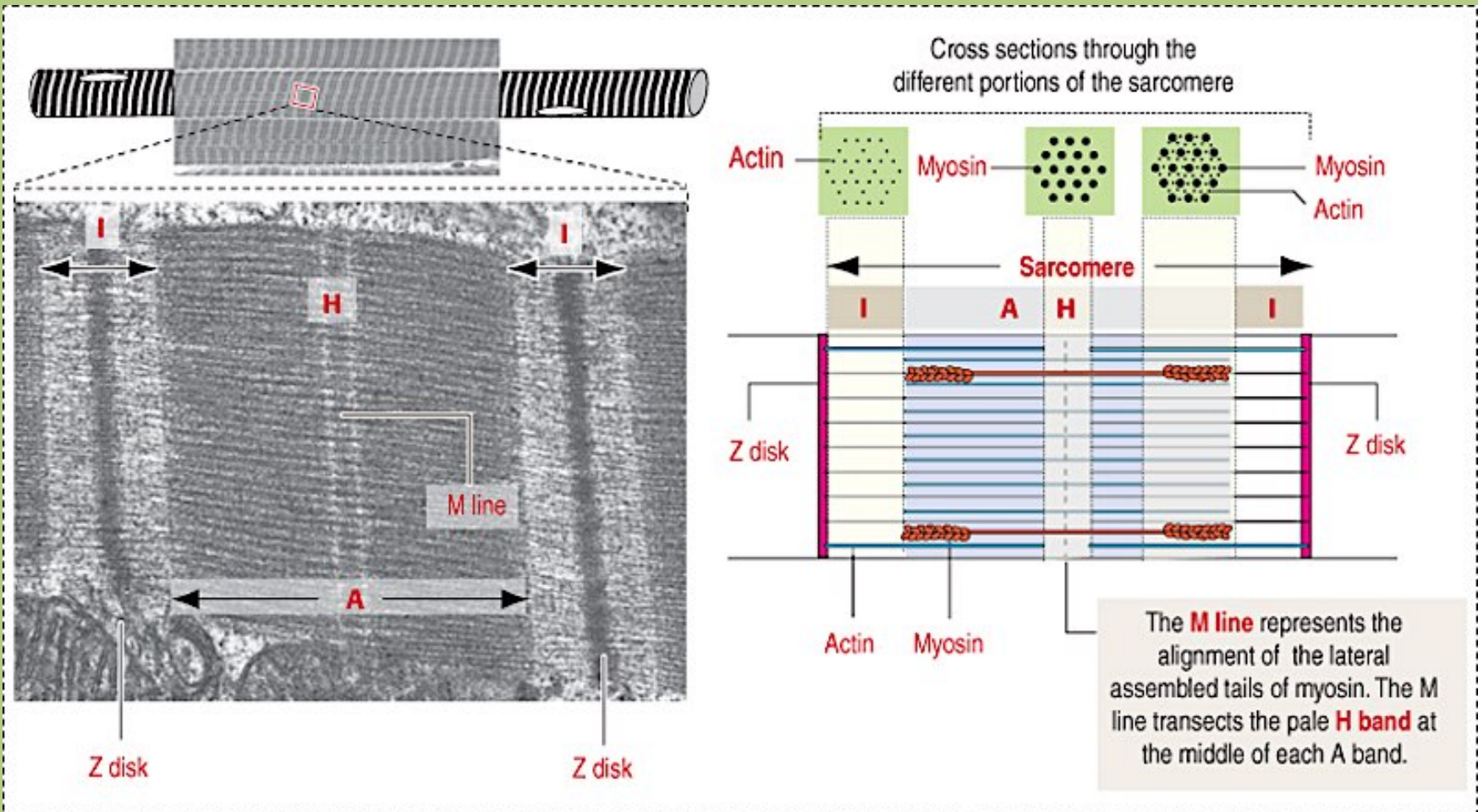
Z line: actin filaments

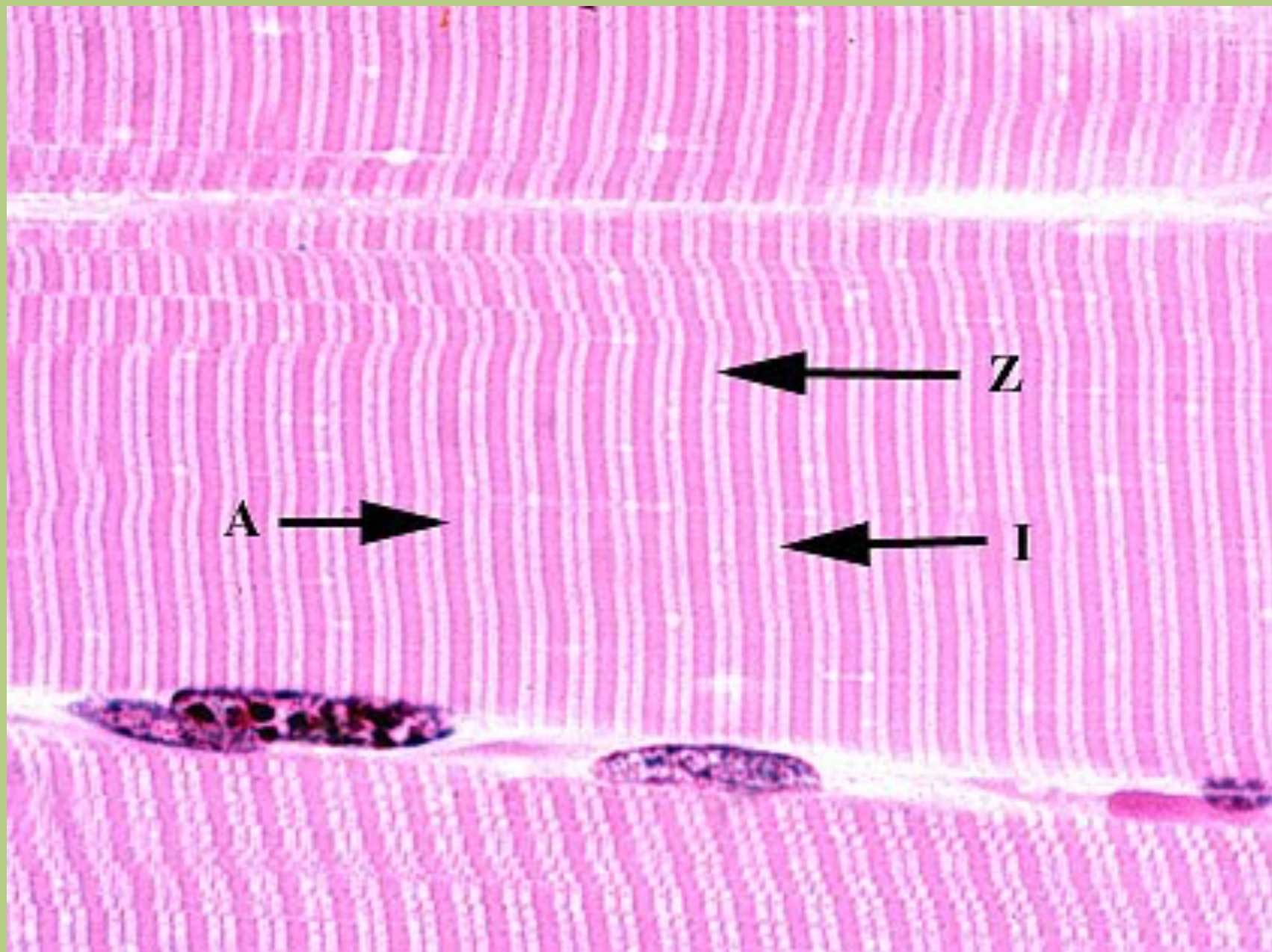
connect to it

H band: only myosin

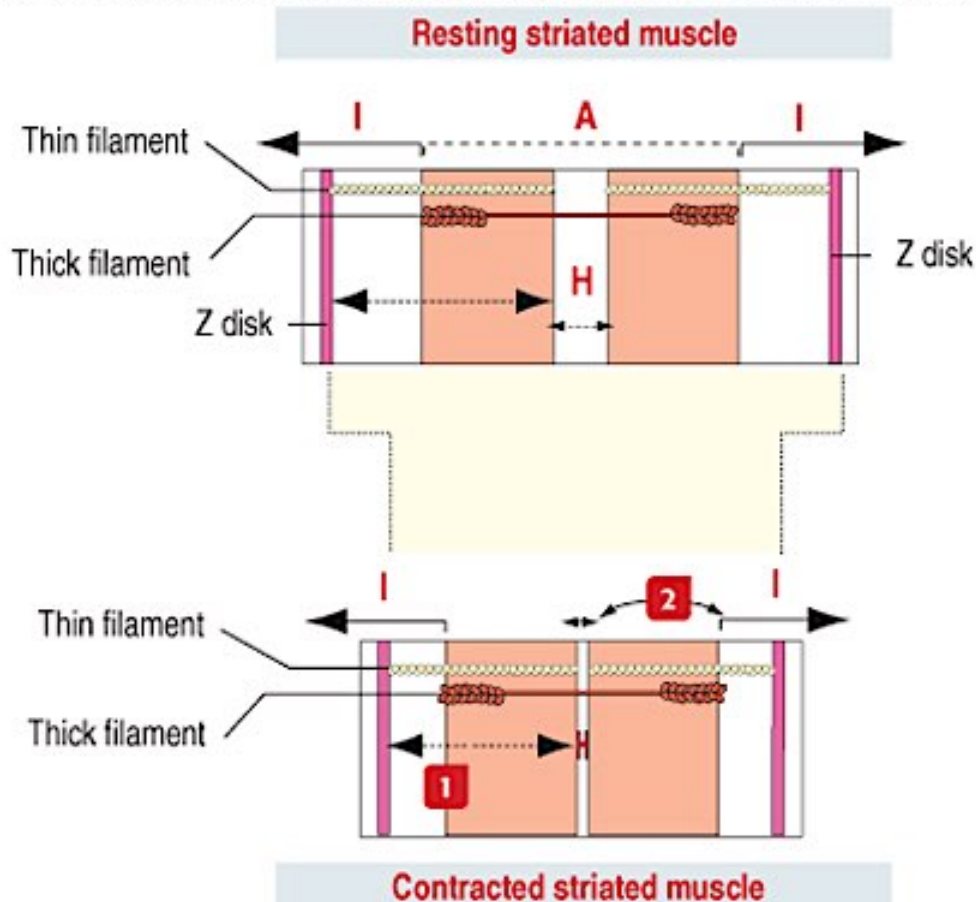
M line: myosin binding C proteins

Sarcomere: unit of the muscle tissue;
distance between two Z lines





SLIDING FILAMENT THEORY



The A band represents the distribution of the myosin thick filaments. The H band represents the myosin tail regions of the thick filaments not overlapping with thin filaments.

Thin filaments are attached to the Z disk. Two half-I bands, containing thin filaments, are seen at the right and left side of the Z disk.

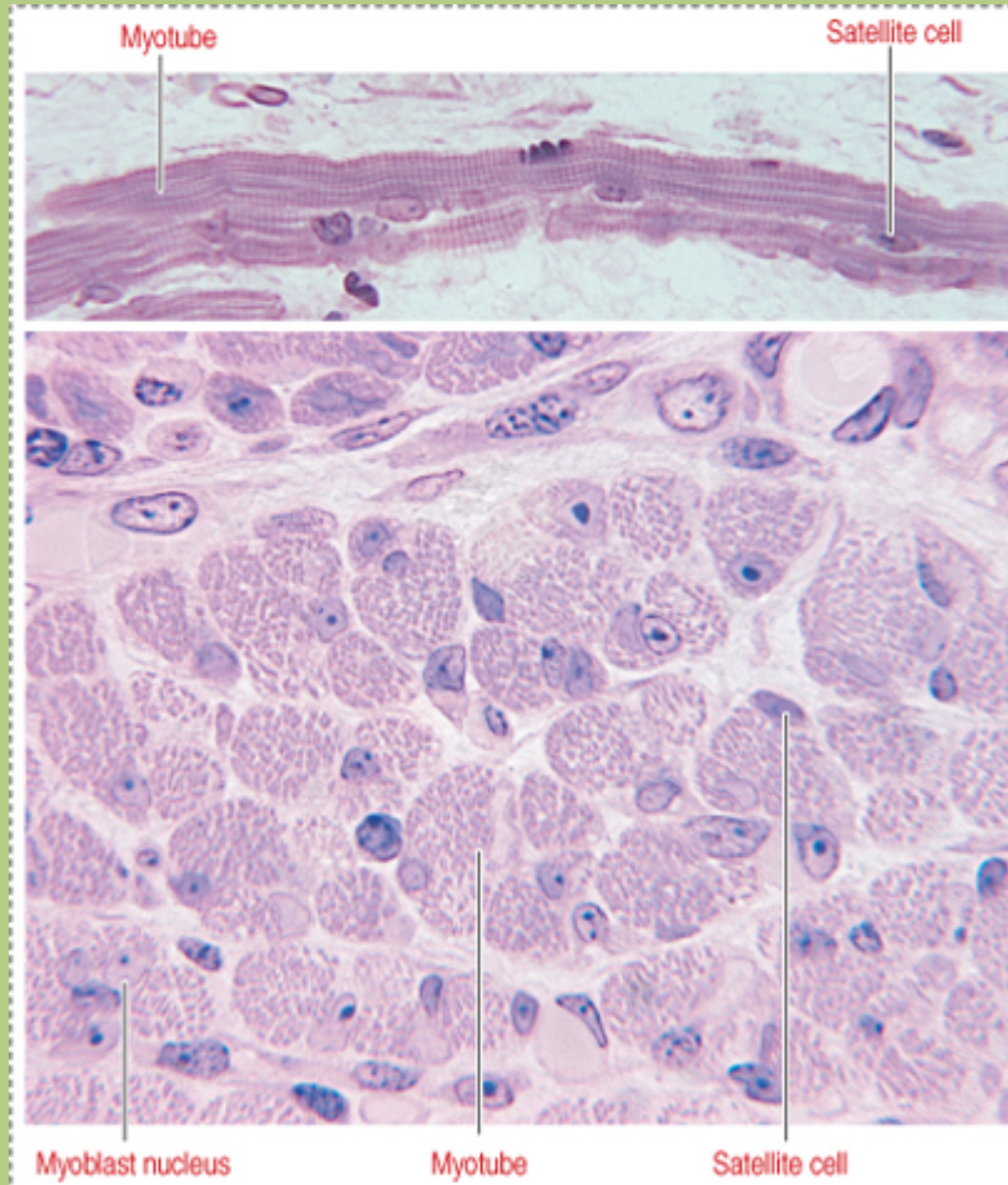
During muscle contraction

1 The length of the thick and thin filaments does not change. This is demonstrated by **the constant length of the A band and the distance between the Z band and the adjacent edge of the H band.**

2 The length of the sarcomere decreases because thick and thin filaments slide past each other. This is demonstrated by **a reduction in the length of the H band and the I band.**

REGENERATION OF STRIATED MUSCLE

- Resident muscle stem cells (=satellite cells), which upon injury activate, proliferate and fuse to repair damaged fibers.

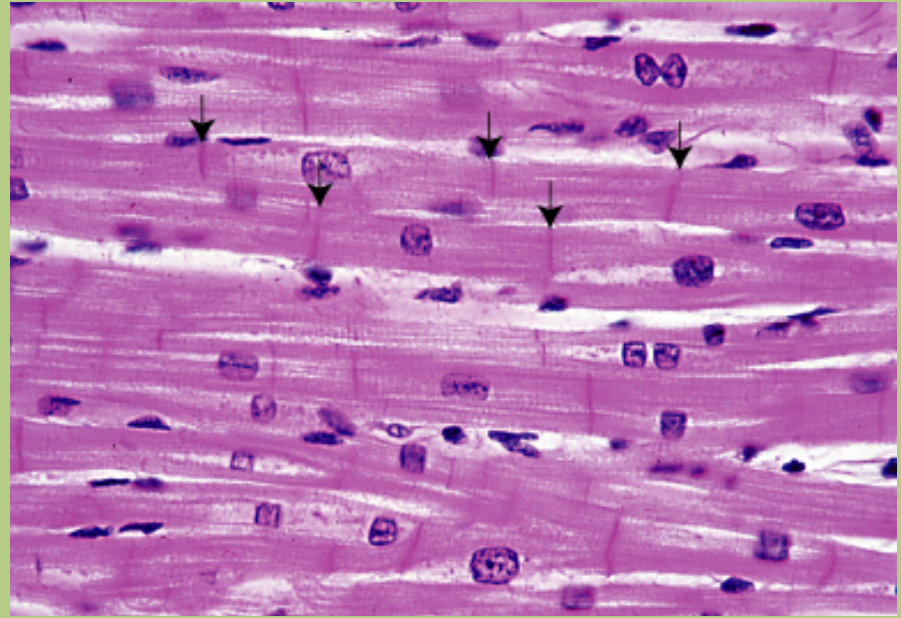


CARDIAC MUSCLE CELLS

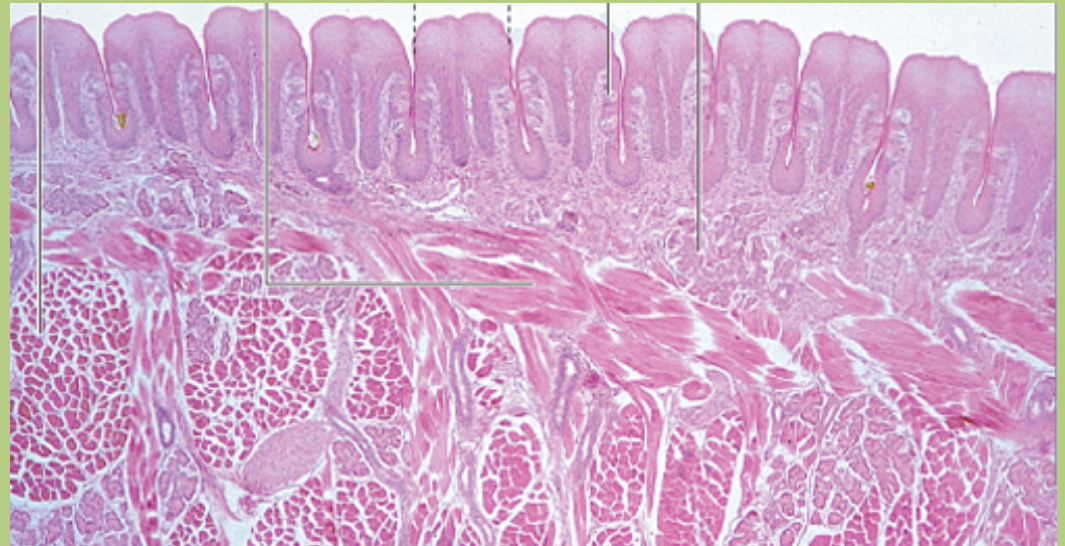
- One nucleus at the center
- Lot of mitochondria
- Actin and myosin filaments
- Well developed cell adhesion complexes (*discus intercalaris* = *Eberth's line*: *fasciae adherentes*; *macula adherens*; *gap junctions*): fast conduction
- Long-lived cells
- No regeneration

TYPES OF STRIATED MUSCLE II.

- *Cardiac muscle*
(arrows: Eberth's lines)



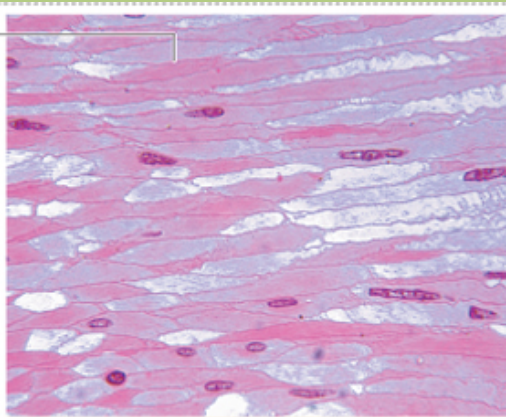
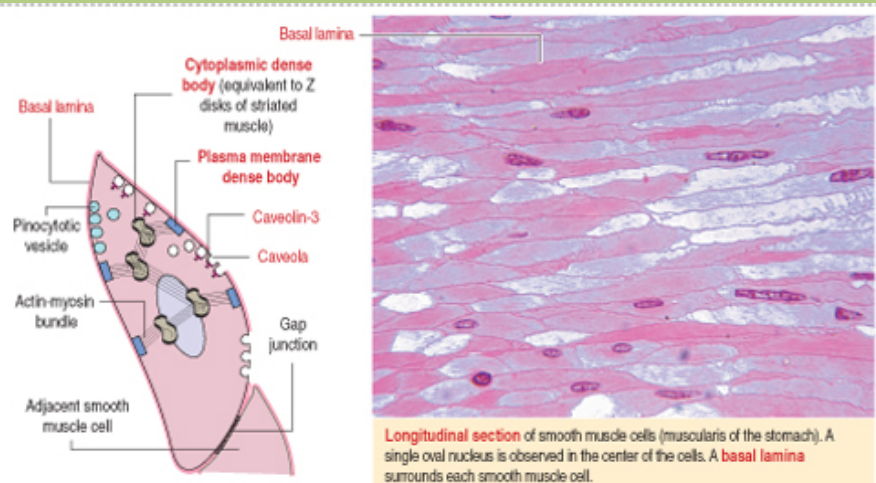
- *Visceral striated muscle*
(tongue)



SMOOTH MUSCLE

- Spindle-shaped cells
- Length from 20 to 200 μm
- One nucleus at the center
- The smooth muscle cells are able to synthesize elastin, collagen and other extracellular matrix's components
- No striation
- Regenerable

SMOOTH MUSCLE



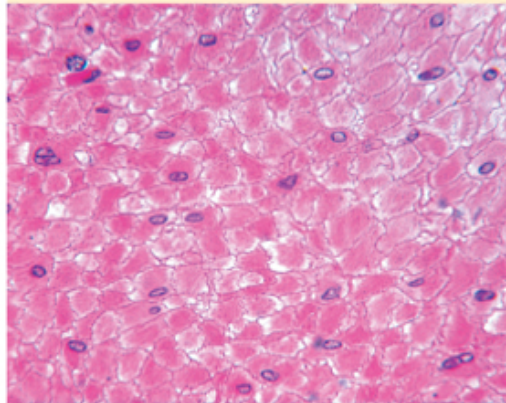
Longitudinal section of smooth muscle cells (muscularis of the stomach). A single oval nucleus is observed in the center of the cells. A basal lamina surrounds each smooth muscle cell.

Characteristics of smooth muscle

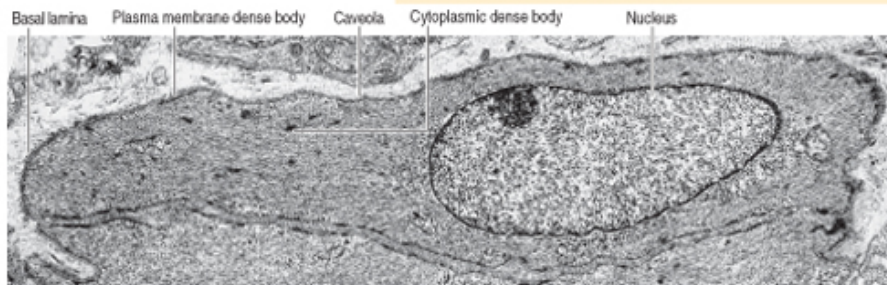
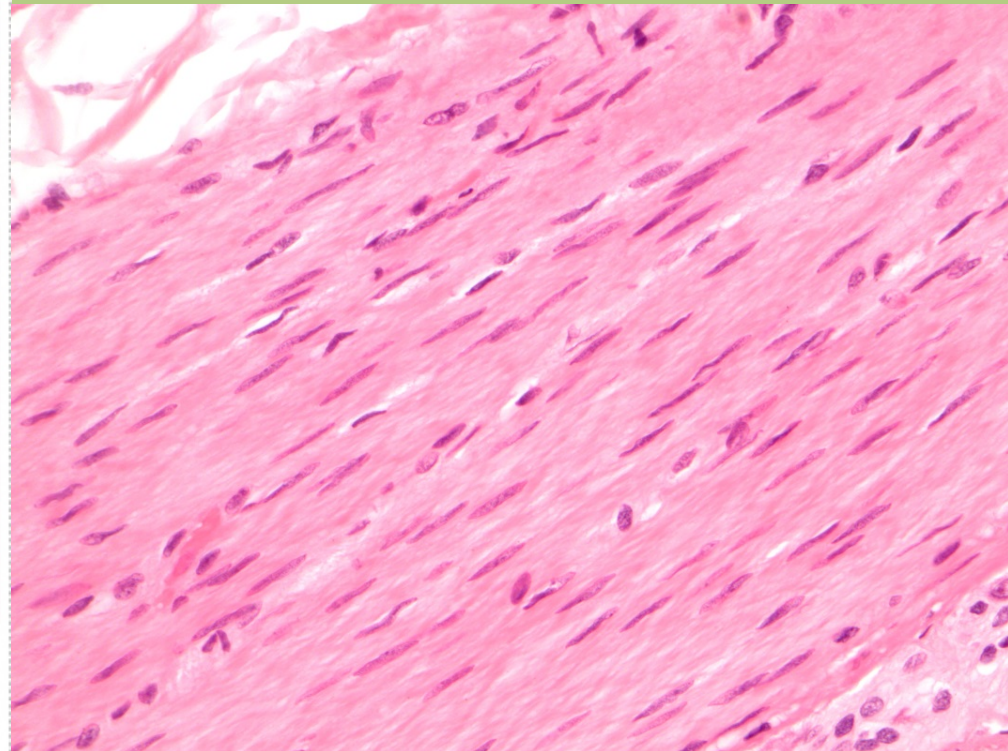
Smooth muscle is found in the walls of tubular organs, the walls of most **blood vessels**, the **iris** and **ciliary body** (eye), and **arrector pili muscle** (hair follicles), among other sites. It consists of fusiform individual cells or fibers with a **central nucleus**. Smooth cells in the walls of large blood vessels produce **elastin**.

Caveolae—depressions of the plasma membrane—are permanent structures involved in fluid and electrolyte transport (**pinocytosis**).

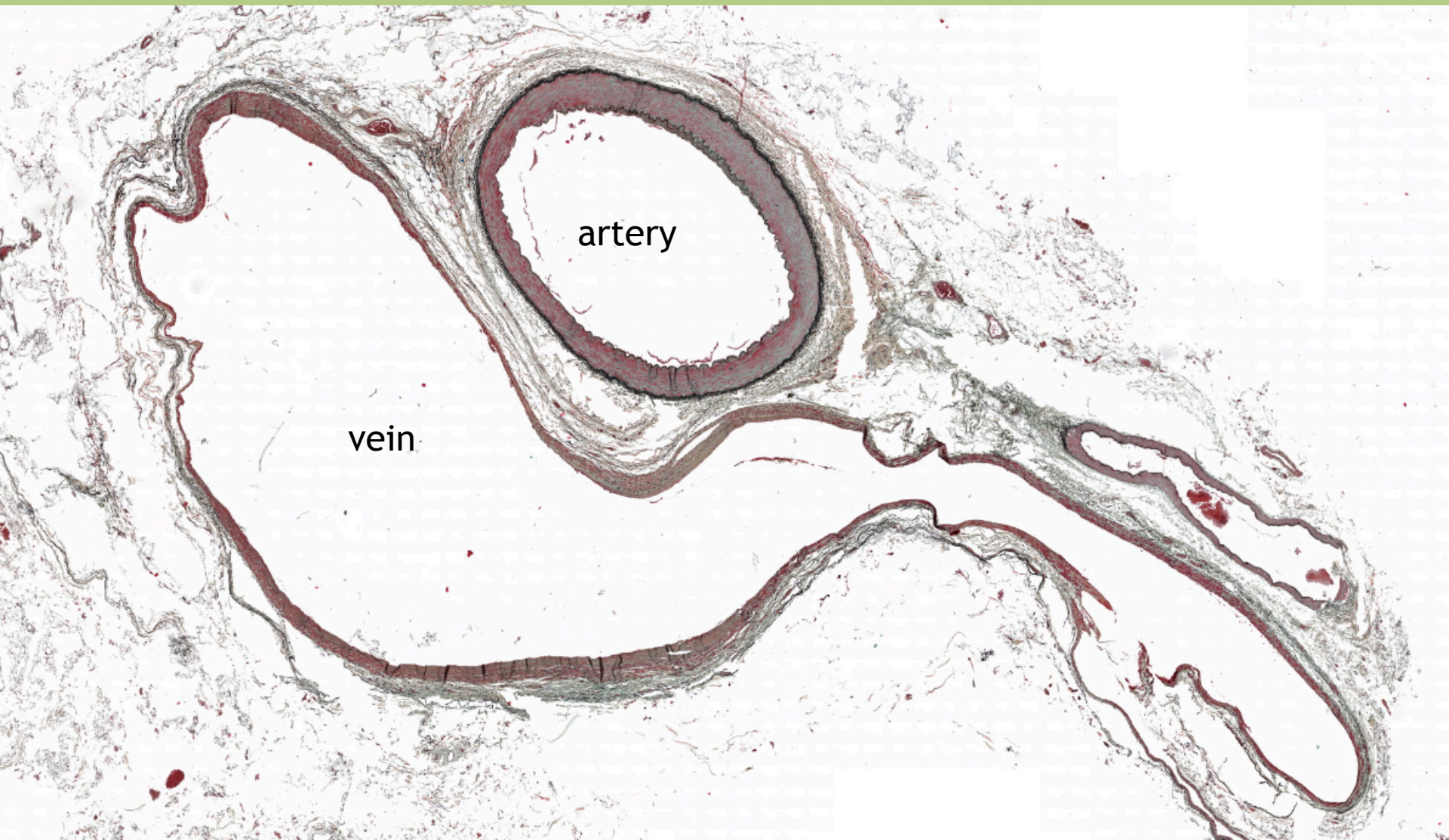
Caveolin-3, a protein encoded by a member of the caveolin gene family, is associated with **lipid rafts**. Complexes formed by caveolin-3 bound to **cholesterol** in a lipid raft invaginate and form caveolae. Caveolae detach from the plasma membrane to form **pinocytotic vesicles**.



Cross section of smooth muscle cells. Depending on the section level, a central nucleus is observed in some of the muscle cells.



Blood vessels



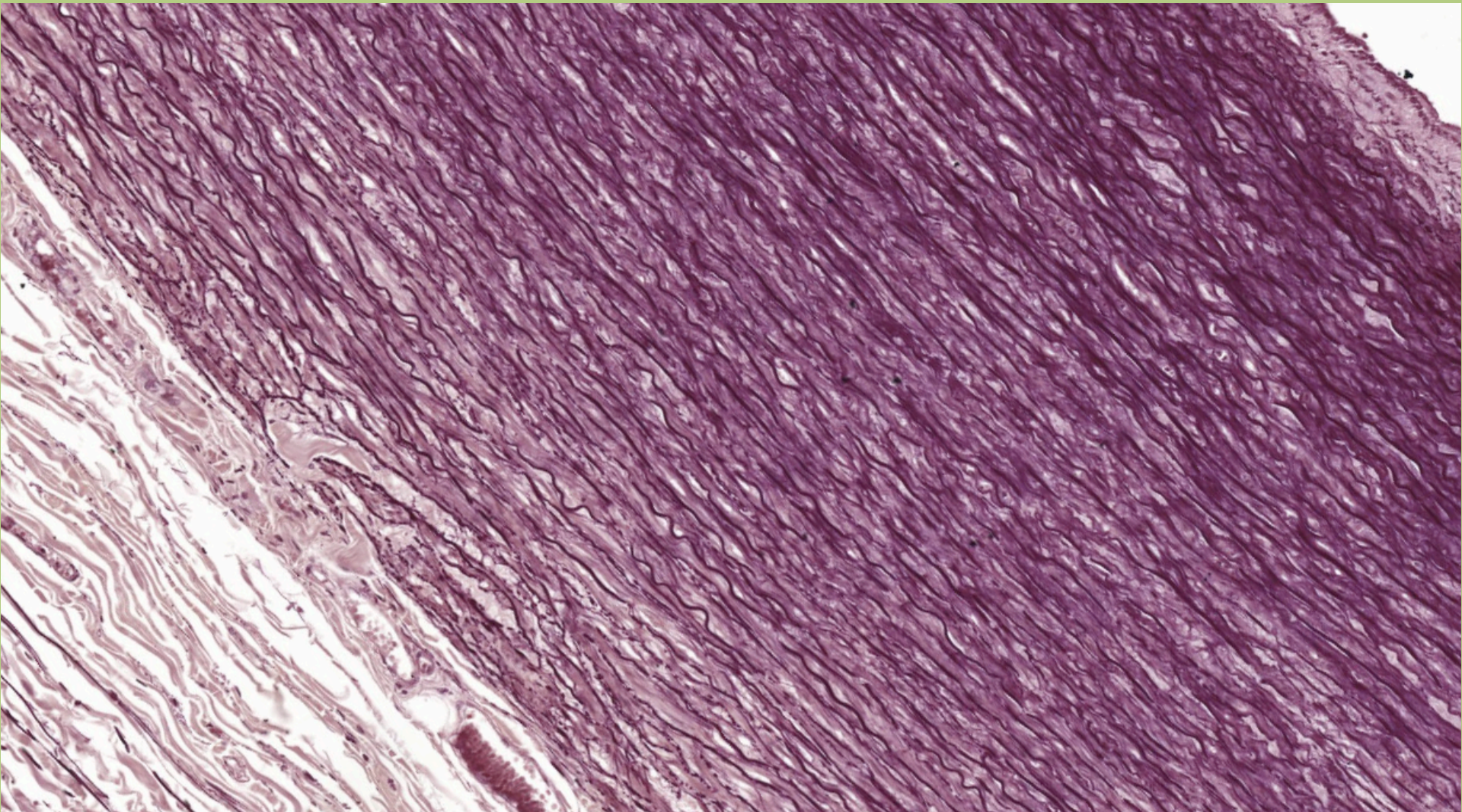
Layers of vascular wall

Tunica intima: *endothelial cells*

Tunica media: *circumferentially arranged layers of smooth muscles cells.* In arteries, this layer is relatively thick.

Tunica adventitia: *outermost connective tissue layer.* In veins, it is quite thick.

Types of vessels: elastic artery

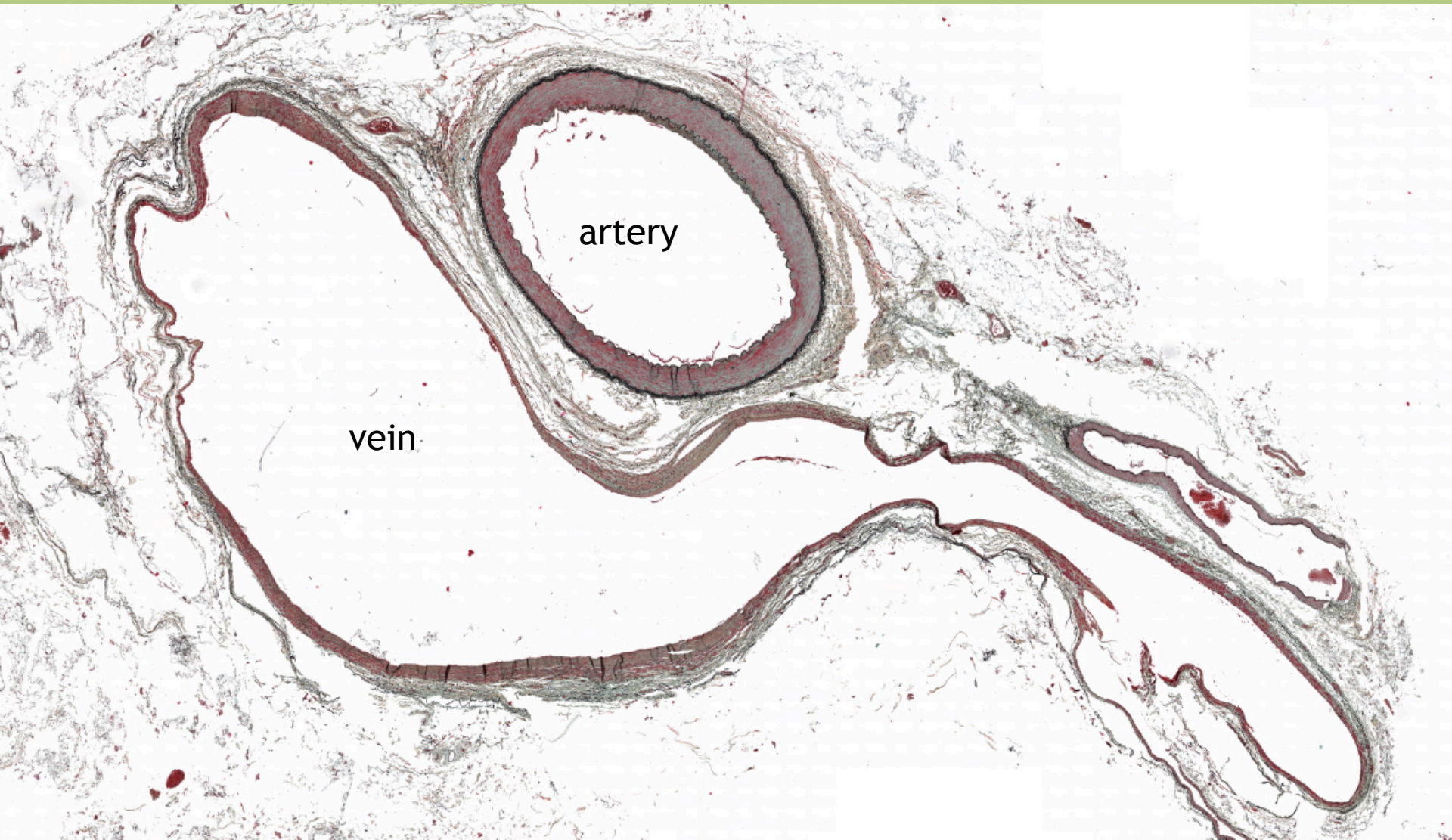


Aorta: the main and largest artery of the human body. Its tunica media is built up by elastic lamellae and smooth muscles cells.

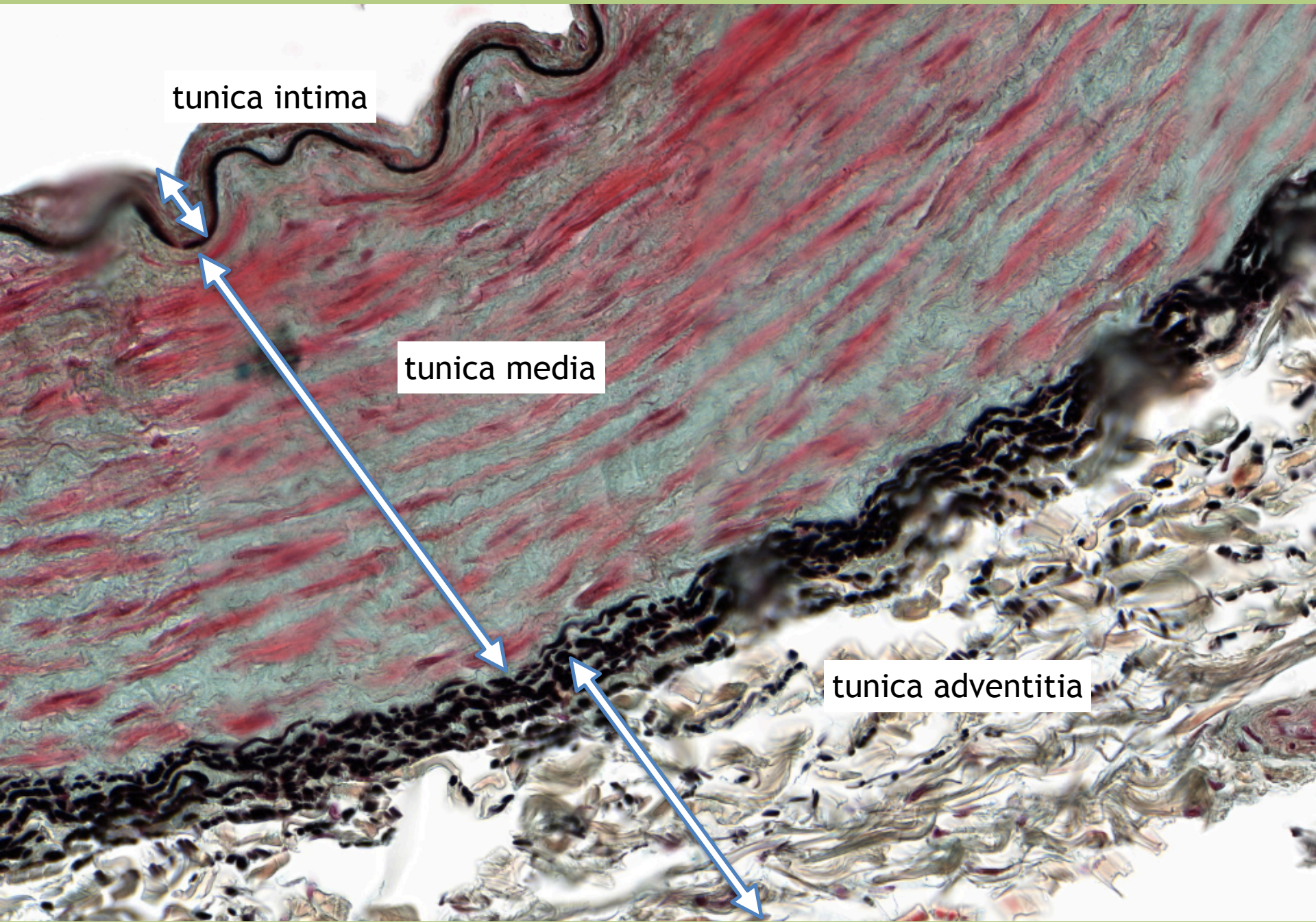
Internal elastic membrane

Vasa vasorum

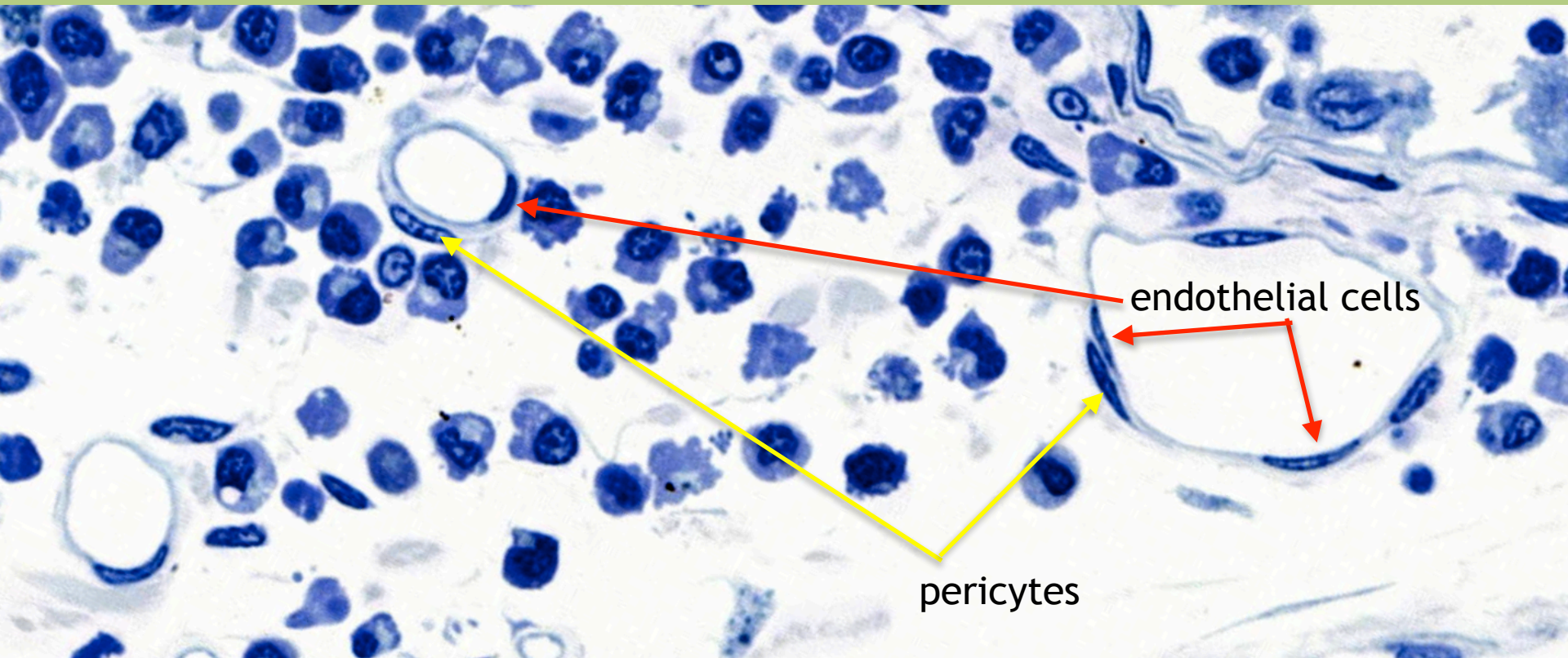
Muscular artery



Muscular artery



Types of vessels: capillaries



Pericytes: they are contractile cells and provide support for capillaries and postcapillary venules. Histologically, pericytes display features of undifferentiated mesenchymal stem cells.

Thank you for your
attention.

References:

McGraw-Hill Company's pictures

Gray's Anatomy for Students

Pearson Education